Assignment 5: Data Visualization

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on Data Visualization

Directions

1. Rename this file `<FirstLast>_A05_DataVisualization.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change "Student Name" on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure your code is tidy; use line breaks to ensure your code fits in the knitted output.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up your session

1. Set up your session. Load the tidyverse, here & cowplot packages, and verify your home directory. Read in the NTL-LTER processed data files for nutrients and chemistry/physics for Peter and Paul Lakes (use the tidy `NTL-LTER_Lake_Chemistry_Nutrients_PeterPaul_Processed.csv` version in the Processed_KEY folder) and the processed data file for the Niwot Ridge litter dataset (use the `NEON_NIWO_Litter_mass_trap_Processed.csv` version, again from the Processed_KEY folder).

2. Make sure R is reading dates as date format; if not change the format to date.

```
#1
knitr::opts_chunk$set(tidy = TRUE, tidy.opts = list(width.cutoff = 20))
library(cowplot)
library(here)
library(tidyverse)
getwd()
```

```
## [1] "C:/Users/24169/Desktop/e872/872_spring26"
```

```
lake <- read.csv(here(
'./Data/Processed_KEY/NTL-LTER_Lake_Chemistry_Nutrients_PeterPaul_Processed.csv'),
stringsAsFactors = TRUE)
litter <- read.csv(here(
```

```r
'./Data/Processed_KEY/NEON_NIWO_Litter_mass_trap_Processed.csv'),
stringsAsFactors = TRUE)
#2
class(lake$sampledate)
```

```
## [1] "factor"
```

```r
class(litter$collectDate)
```

```
## [1] "factor"
```

```r
lake <- lake |> mutate(sampledate = ymd(sampledate))
litter <- litter |> mutate(collectDate = ymd(collectDate))
class(lake$sampledate)
```

```
## [1] "Date"
```

```r
class(litter$collectDate)
```

```
## [1] "Date"
```

Define your theme

3. Build a theme and set it as your default theme. Customize the look of at least two of the following:

- Plot background
- Plot title
- Axis labels
- Axis ticks/gridlines
- Legend

```r
# 3
design <- theme_minimal() +
    theme(plot.background = element_rect(fill = "#F0FFFF"),
        plot.title = element_text(color = "#312C85"),
        plot.subtitle = element_text(color = "#193CB8"),
        axis.text = element_text(face = "bold"),
        panel.grid = element_line(color = "#000000"),
        legend.background = element_rect(fill = "#90EE90"),
        legend.title = element_text(color = "#721378"),
        legend.position = "bottom")
theme_set(design)
```

Create graphs

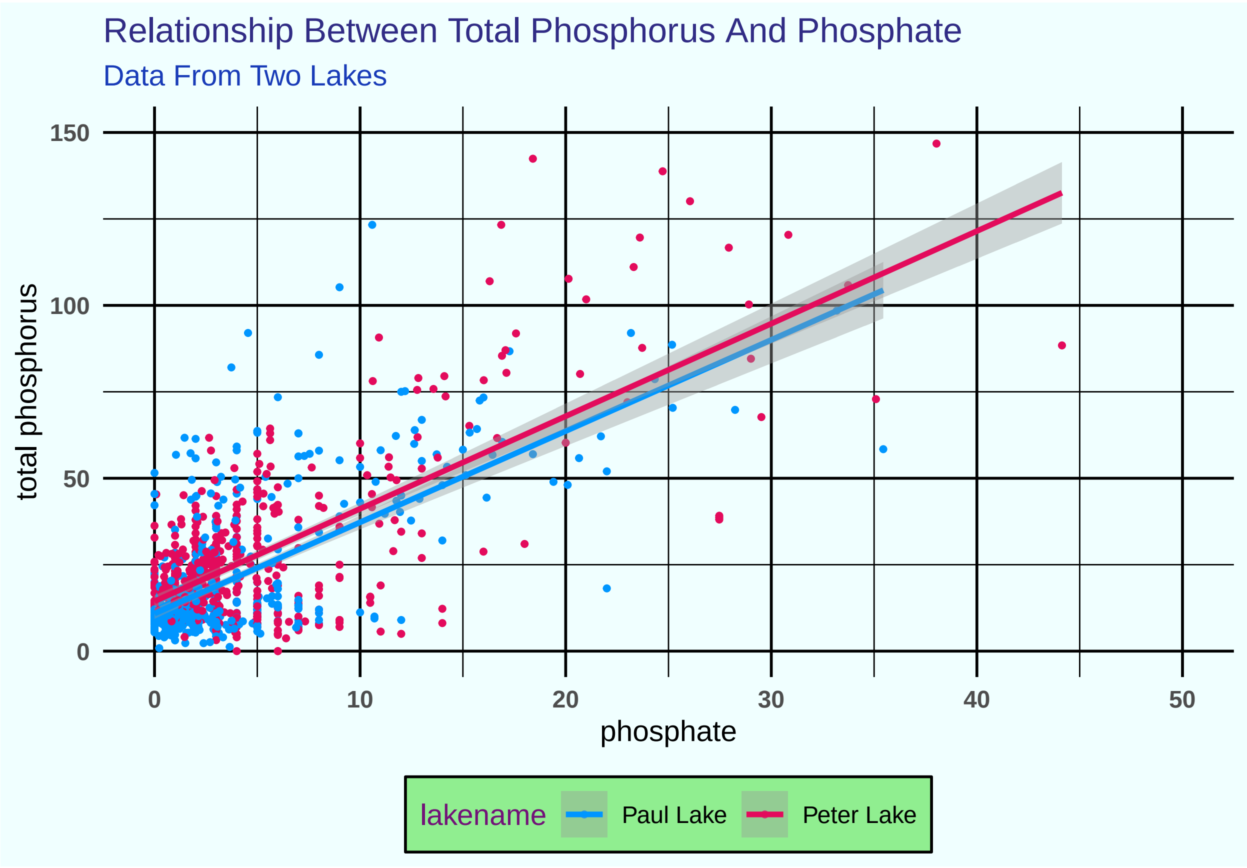
For numbers 4-7, create ggplot graphs and adjust aesthetics to follow best practices for data visualization. Ensure your theme, color palettes, axes, and additional aesthetics are edited accordingly.

4. [NTL-LTER] Plot total phosphorus (`tp_ug`) by phosphate (`po4`), with separate aesthetics for Peter and Paul lakes. Add line(s) of best fit using the `lm` method. Adjust your axes to hide extreme values (hint: change the limits using `xlim()` and/or `ylim()`).

```
# 4
length(unique(lake$lakename))
```

```
## [1] 2
```

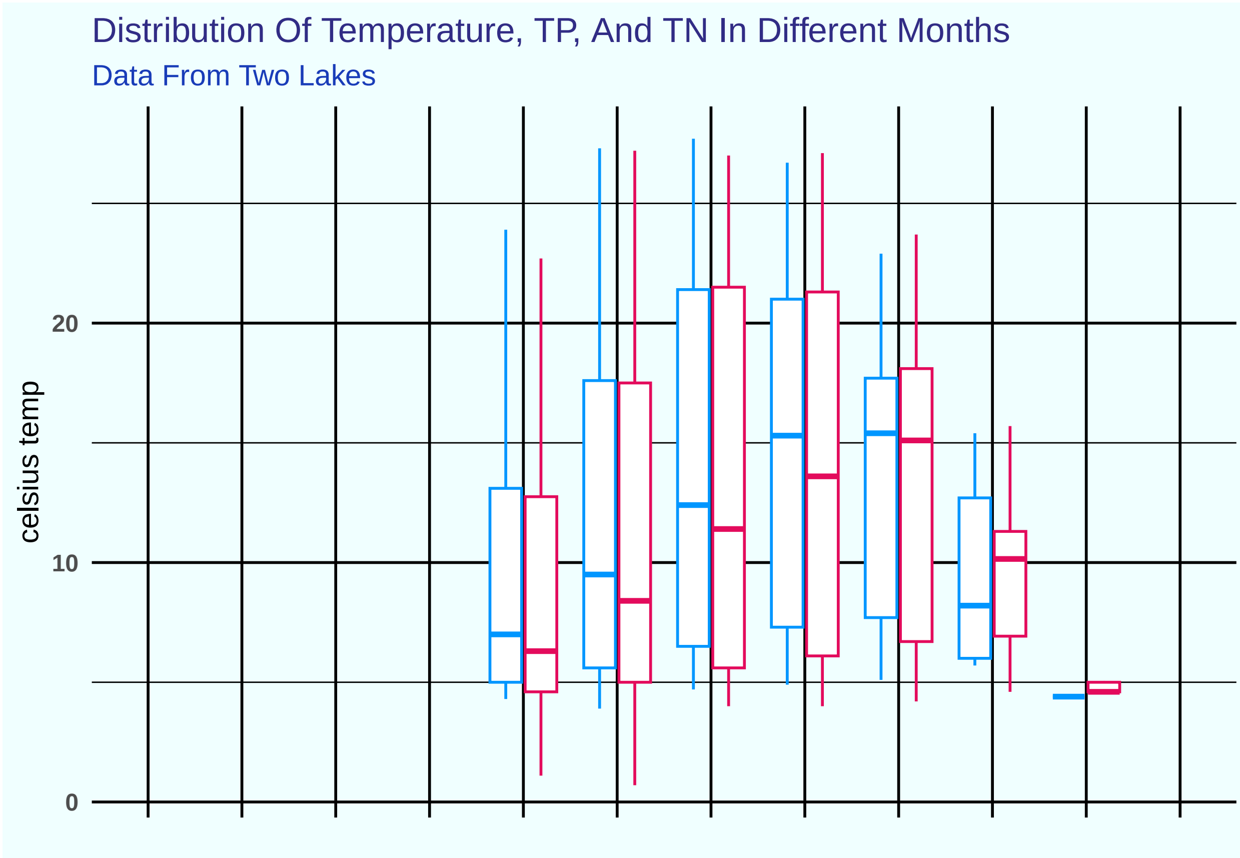
```
p1 <- ggplot(lake, aes(x = po4,
    y = tp_ug, color = lakename)) +
    geom_point(size = 0.75,
        na.rm = TRUE) +
    geom_smooth(linewidth = 1,
        method = lm,
        na.rm = TRUE) +
    scale_color_manual(values = c("#0096FF",
        "#E30B5C")) +
    xlim(0, 50) + ylim(0,
    150) + labs(title = "Relationship Between Total Phosphorus And Phosphate",
    subtitle = "Data From Two Lakes",
    x = "phosphate",
    y = "total phosphorus")
p1
```



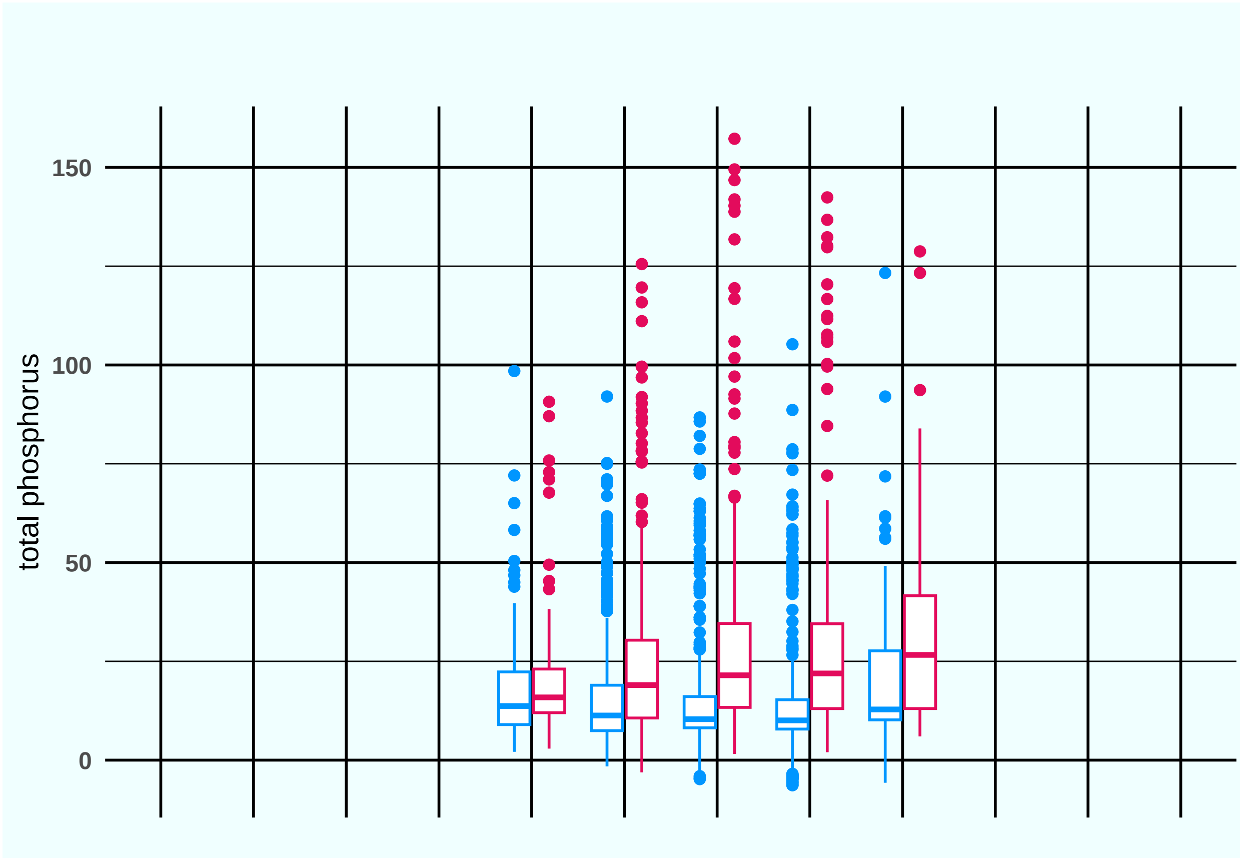
5. [NTL-LTER] Make three separate boxplots of (a) temperature, (b) TP, and (c) TN, with month as the x axis and lake as a color aesthetic. Then, create a cowplot that combines the three graphs. Make sure that only one legend is present and that graph axes are aligned. Show all months, even those where no data was collected.

Tips: * Recall the discussion on factors in the lab section as it may be helpful here. * Setting an axis text in your theme to `element_blank()` removes the axis text (useful when multiple, aligned plots use the same axis values) * Setting a legend's position to "none" will remove the legend from a plot. * Individual plots can have different relative sizes when combined using `cowplot`.

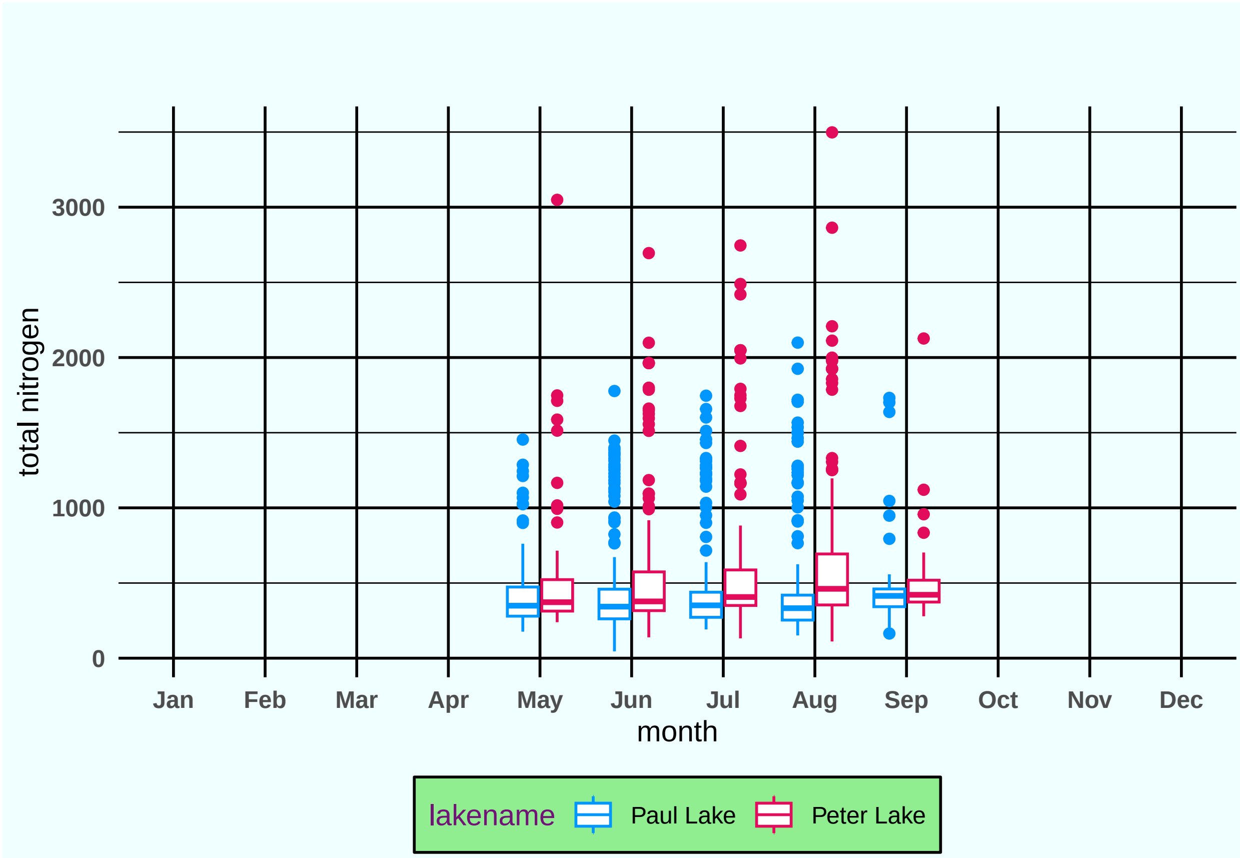
```
# 5
temperature <- ggplot(lake,
    aes(x = factor(month,
        levels = 1:12,
        labels = month.abb),
        y = temperature_C,
        color = lakename)) +
    geom_boxplot(na.rm = TRUE) +
    scale_color_manual(values = c("#0096FF",
        "#E30B5C")) +
    scale_x_discrete(drop = FALSE) +
    theme(axis.text.x = element_blank(),
        legend.position = "none") +
    labs(title = "Distribution Of Temperature, TP, And TN In Different Months",
        subtitle = "Data From Two Lakes",
        x = "", y = "celsius temp")
temperature
```



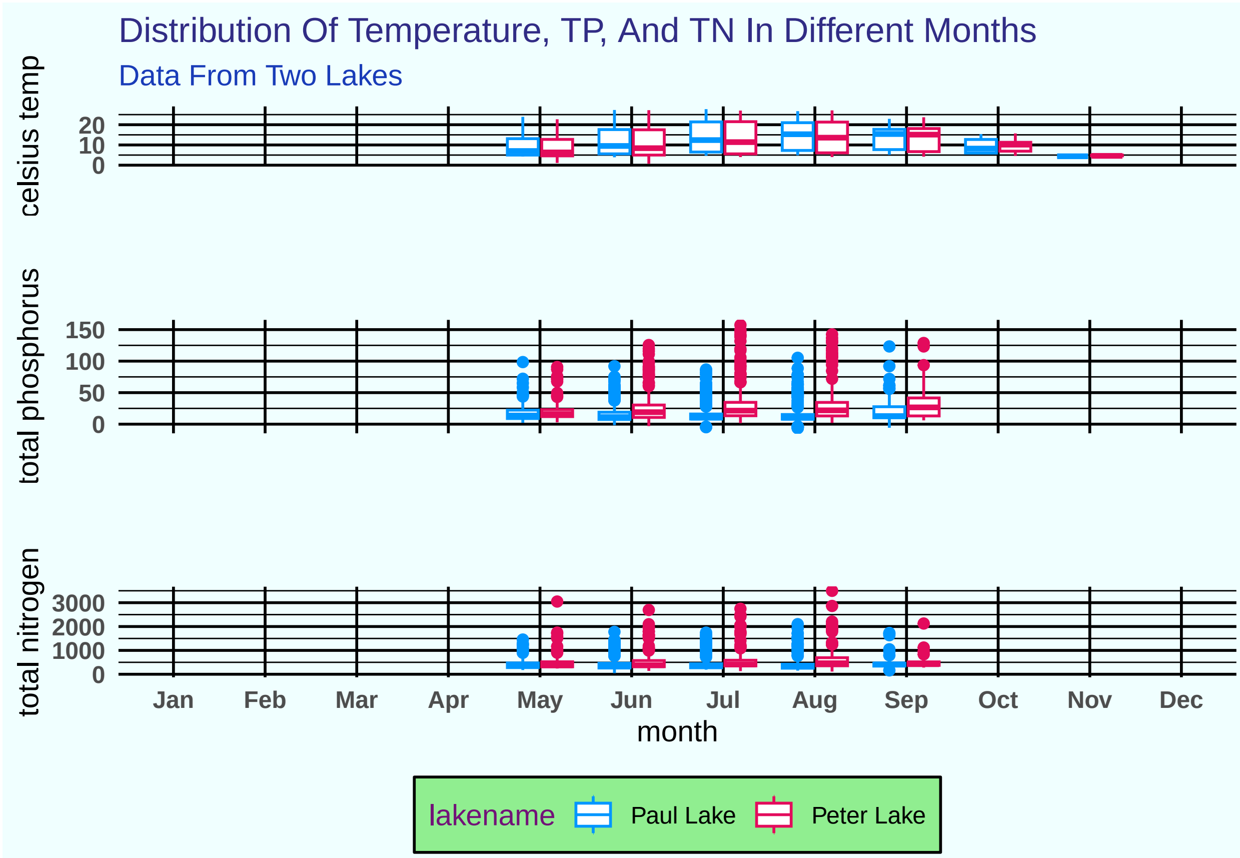

```
total_P <- ggplot(lake,
    aes(x = factor(month,
        levels = 1:12,
        labels = month.abb),
        y = tp_ug, color = lakename)) +
    geom_boxplot(na.rm = TRUE) +
    scale_color_manual(values = c("#0096FF",
        "#E30B5C")) +
    scale_x_discrete(drop = FALSE) +
    theme(axis.text.x = element_blank(),
        legend.position = "none") +
    labs(title = "",
        subtitle = "",
        x = "", y = "total phosphorus")
total_P
```




```r
total_N <- ggplot(lake,
    aes(x = factor(month,
        levels = 1:12,
        labels = month.abb),
        y = tn_ug, color = lakename)) +
    geom_boxplot(na.rm = TRUE) +
    scale_color_manual(values = c("#0096FF",
        "#E30B5C")) +
    scale_x_discrete(drop = FALSE) +
    labs(title = "",
        subtitle = "",
        x = "month",
        y = "total nitrogen")
total_N
```




```
p2 <- plot_grid(temperature,
    total_P, total_N,
    nrow = 3, align = "v",
    rel_heights = c(1,
        1.25, 1.8))
p2
```



Question: What do you observe about the variables of interest over seasons and between lakes?

> Answer: For both lakes, the highest median temperature occurs in September. Before October, the median temperature for Paul Lake is higher than that for Peter Lake. The median TP for Peter Lake is always higher than that for Paul Lake. Between May and September, while median TP for Peter Lake increases continually, there is an approximate quadratic tendency for Paul Lake. The median TN for Peter Lake is always higher than that for Paul Lake. While August features the minimum median TN for Paul Lake, it features the maximum median TN for Peter Lake.

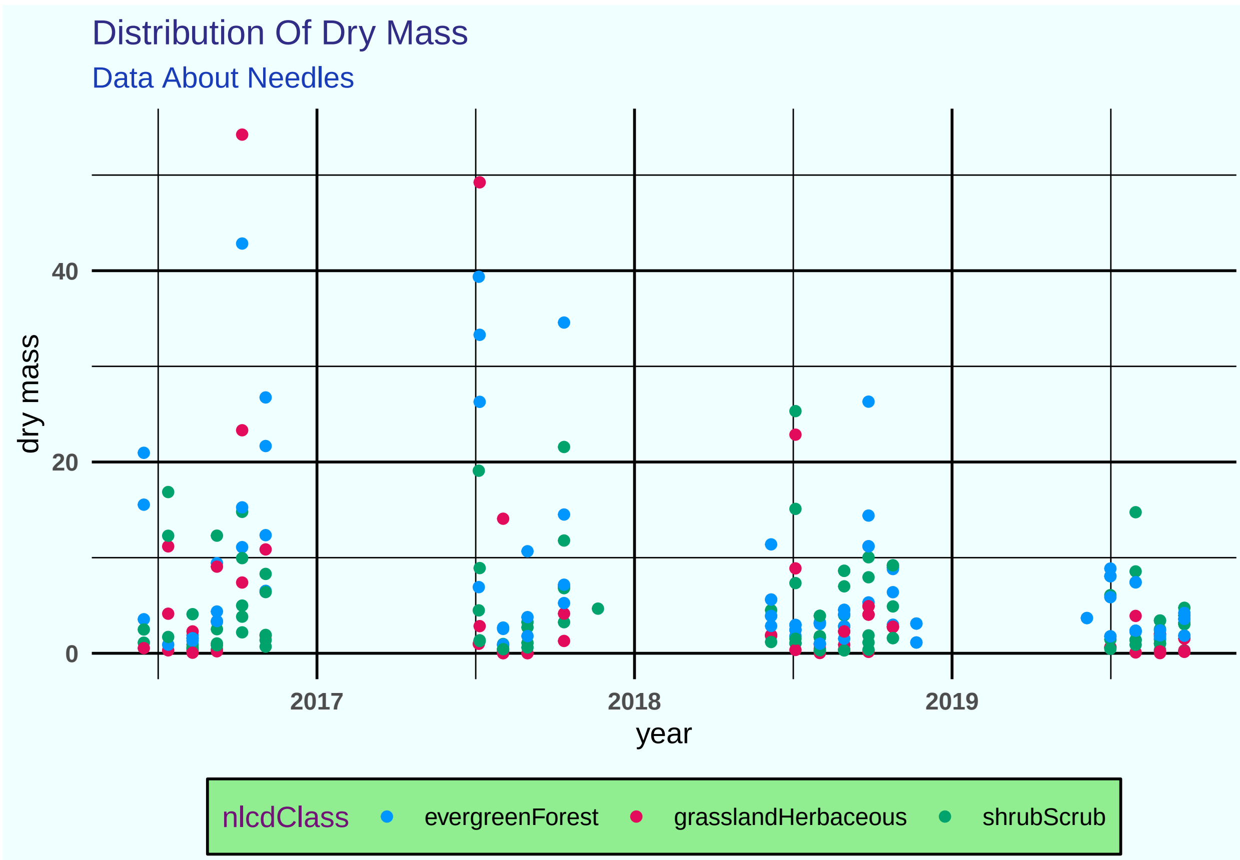
6. [Niwot Ridge] Plot a subset of the litter dataset by displaying only the "Needles" functional group. Plot the dry mass of needle litter by date and separate by NLCD class with a color aesthetic. (no need to adjust the name of each land use)
7. [Niwot Ridge] Now, plot the same plot but with NLCD classes separated into three facets rather than separated by color.

```
# 6
length(unique(litter$nlcdClass))
```

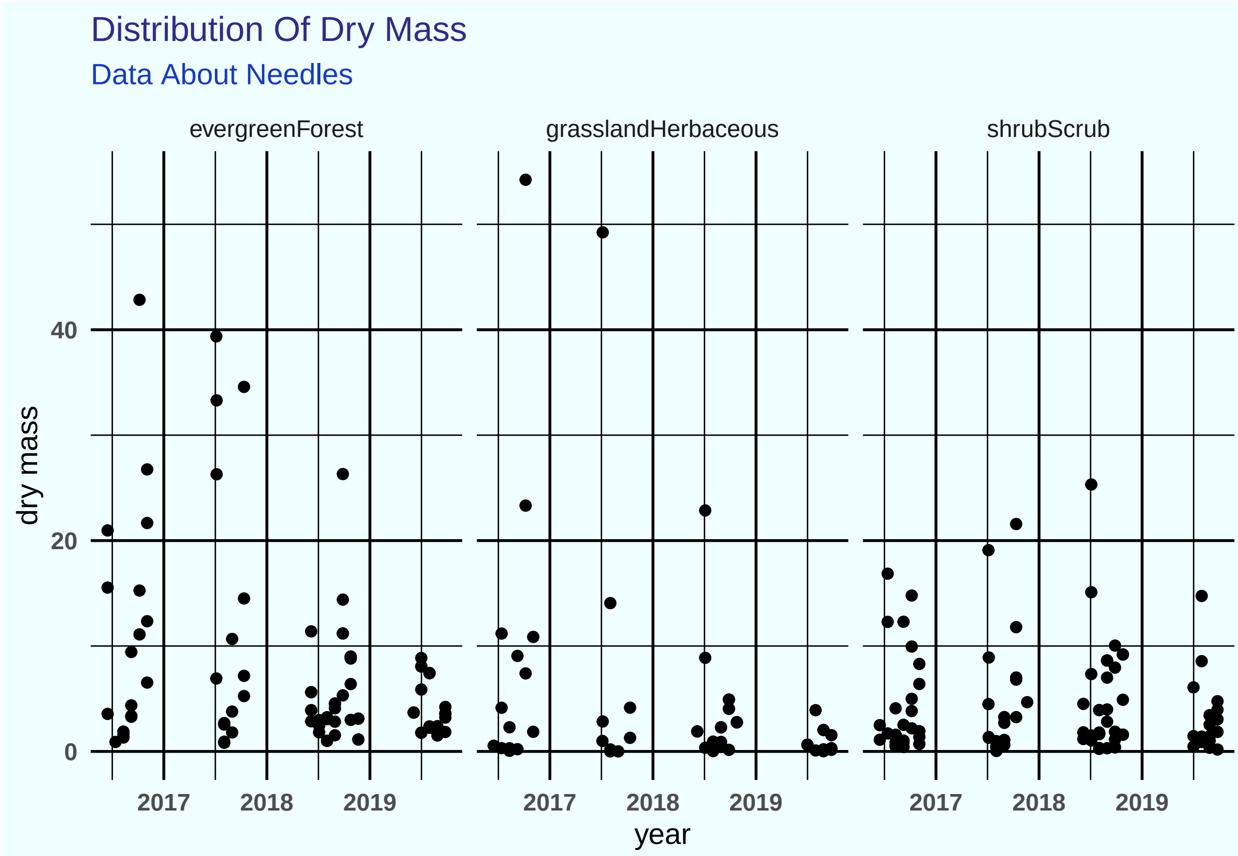
```
## [1] 3
```

```
p3 <- ggplot(subset(litter,
    functionalGroup ==
        "Needles"), aes(x = collectDate,
```

```
    y = dryMass, color = nlcdClass)) +
    geom_point(size = 1.5,
        na.rm = TRUE) +
    scale_color_manual(values = c("#0096FF",
        "#E30B5C", "#00A36C")) +
    labs(title = "Distribution Of Dry Mass",
        subtitle = "Data About Needles",
        x = "year", y = "dry mass")
p3
```




```
# 7
p4 <- ggplot(subset(litter,
    functionalGroup ==
        "Needles"), aes(x = collectDate,
    y = dryMass)) + geom_point(size = 1.5,
    na.rm = TRUE) + facet_grid(~nlcdClass) +
    labs(title = "Distribution Of Dry Mass",
        subtitle = "Data About Needles",
        x = "year", y = "dry mass")
p4
```



Question: Which of these plots (6 vs. 7) do you think is more effective, and why?

Answer: The plot in Question 7 is more effective. The three facets clearly indicate different NLCD classes, facilitating the comparison. Conversely, using a color palette to differentiate in one plot is kind of obscure.